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Batch: A-2
Div: A

Subject: Programming with Java

ASSIGNMENT No : 4

TITLE : Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp4/Main.java
Wrapped and Unwrapped variables :

Wrapped Int : 10
Unwrapped Int : 10

Wrapped Char : A
Unwrapped Char : A
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp4/Exercise/Ex1.java
Percentage : 97.0 %
Total Marks : 291.0 / 300
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp4/Exercise/Ex2.java
Enter Employee ID : -1
Employee ID cannot be negative.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp4/Exercise/Ex2.java
Enter Employee ID : 100
Enter Employee Salary : -1
Employee Salary cannot be negative.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp4/Exercise/Ex2.java
Enter Employee ID : 100
Enter Employee Salary : 100000
Enter Employee Bonus : -1
Employee Bonus cannot be negative.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp4/Exercise/Ex2.java
Enter Employee ID : 100
Enter Employee Salary : 100000
Enter Employee Bonus : 50000

Employee Details :
ID          : 100
Salary      : Rs.100000
Bonus       : Rs.50000
Net Salary  : Rs.150000
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the concept and importance of Java wrapper classes for converting primitive data types into objects and vice versa. It demonstrated the use of wrapper classes such as Integer, Double, Character, and Boolean for autoboxing, unboxing, parsing, and basic operations on wrapped values. The implementation improved my understanding of data conversion, input validation, and object-oriented programming concepts. These concepts are widely used in Java Collections Framework, file handling, database applications, and other real-world Java programs where object representation of primitive data is required.